State-space models

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Outline

1. Presentation of state-space models
2. Examples of state-space models
3. Sequential analysis of state-space models

Objectives

The sequential analysis of state-space models is the main (but not only) application of Sequential Monte Carlo.

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The sequential analysis of state-space models is the main (but not only) application of Sequential Monte Carlo.
The aim of this chapter is to define state-space models, give examples of such models from various areas of science, and discuss their main properties.

A first definition (with functions)

A time series model that consists of two discrete-time processes $\{X_t\} := (X_t)_{t \geq 0}$, $\{Y_t\} := (Y_t)_{t \geq 0}$, taking values respectively in spaces $\mathcal{X}$ and $\mathcal{Y}$, such that

$$X_t = K_t(X_{t-1}, U_t, \theta), \quad t \geq 1$$
$$Y_t = H_t(X_t, V_t, \theta), \quad t \geq 0$$

where K_0, K_t, H_t, are determistic functions, $\{U_t\}$, $\{V_t\}$ are sequences of i.i.d. random variables (*noises*, or *shocks*), and $\theta \in \Theta$ is an unknown parameter.

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This is a popular way to define SSMs in Engineering. Rigorous, but not sufficiently general.

A second definition (with densities)

$$
\begin{aligned}
p_\theta(x_0) &= p_0^\theta(x_0) \\
p_\theta(x_t|x_{0:t-1}) &= p_t^\theta(x_t|x_{t-1}) \quad t \geq 1 \\
p_\theta(y_t|x_{0:t}, y_{0:t-1}) &= f_t^\theta(y_t|x_t)
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Not so rigorous (or not general enough): some models are such that $X_t|X_{t-1}$ does not admit a probability density (with respect to a fixed dominating measure).

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Signal processing: tracking, positioning, navigation

X_t is position of a moving object, e.g.

$$X_t = X_{t-1} + U_t, \quad U_t \sim \mathcal{N}_2(0, \sigma^2 I_2),$$

and Y_t is a measurement obtained by e.g. a radar,

$$Y_t = \mathrm{atan}\left(\frac{X_t(2)}{X_t(1)}\right) + V_t, \quad V_t \sim \mathcal{N}_1(0, \sigma_Y^2).$$

and $\theta = (\sigma^2, \sigma_Y^2)$.

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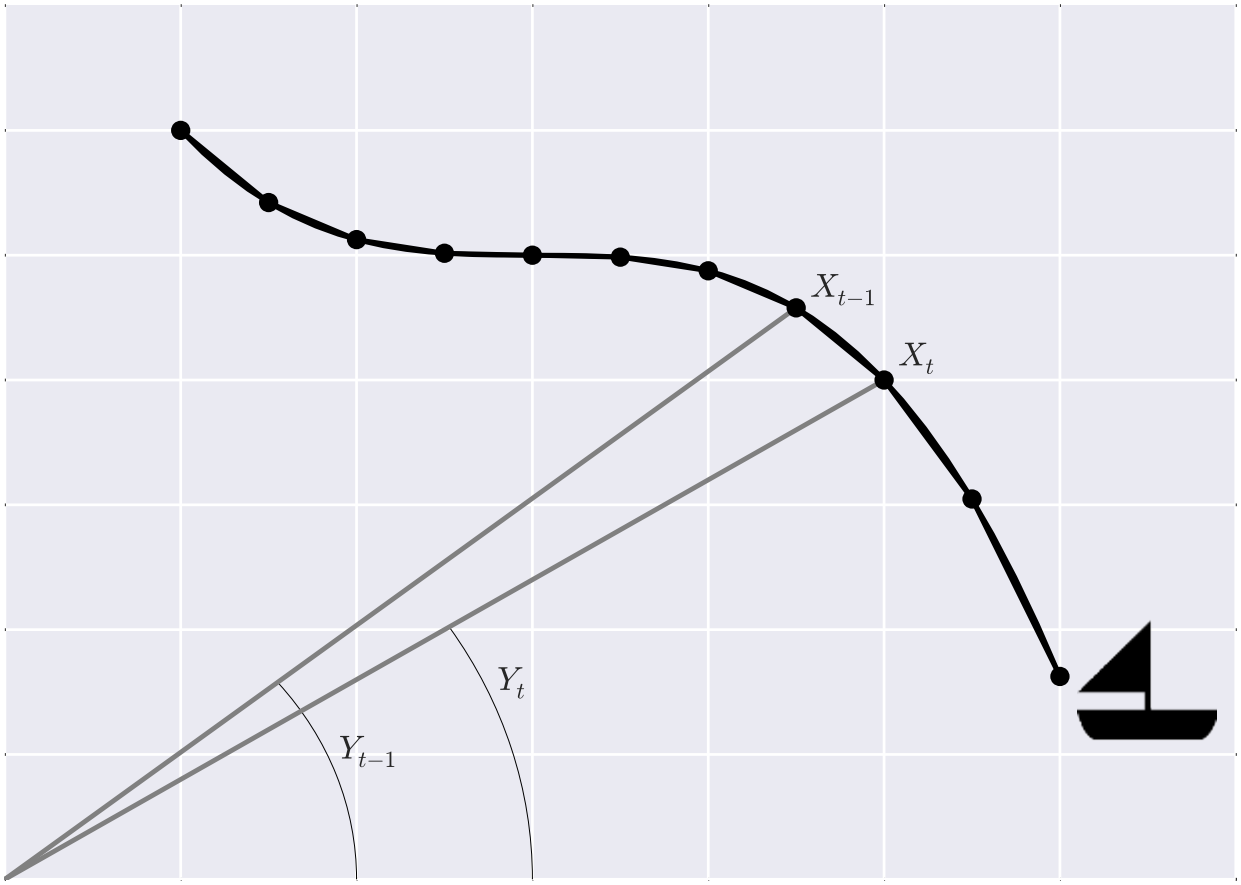
(This is called the **bearings-only tracking** model.)

Plot



GPS

In GPS applications, the velocity v_t of the vehicle is observed, so motion model is (some variation of):

$$X_t = X_{t-1} + v_t + U_t, \qquad U_t \sim \mathcal{N}_2(0, \sigma^2 I_2).$$

Also Y_t usually consists of more than one measurement.

More advanced motion model

A random walk is too erratic for modelling the position of the target; assume instead its velocitity follows a random walk. Then define:

$$X_t = \begin{pmatrix} I_2 & I_2 \\ 0_2 & I_2 \end{pmatrix} X_{t-1} + \begin{pmatrix} 0_2 & 0_2 \\ 0_2 & U_t \end{pmatrix}, \quad U_t \sim \mathcal{N}_2(0, \sigma^2 I_2),$$

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with obvious meanings for matrices 0_2 and I_2.
Note: $X_t(1)$ and $X_t(2)$ (position) are deterministic functions of X_{t-1}: no probability density for $X_t | X_{t-1}$.

multi-target tracking

Same ideas except $\{X_t\}$ now represent the position (and velocity if needed) of a set of targets (of random size); i.e. $\{X_t\}$ is a point process.

Time series of counts (neuro-decoding, astrostatistics, genetics)

- Neuro-decoding: Y_t is a vector of d_y counts (spikes from neuron k),

$$Y_t(k)|X_t \sim \mathcal{P}(\lambda_k(X_t)), \quad \log \lambda_k(X_t) = \alpha_k + \beta_k X_t,$$

  and X_t is position+velocity of the subject's hand (in 3D).
- astro-statistics: Y_t is number of photon emissions; intensity varies over time (according to an auto-regressive process)
- Y_t is the number of 'reads', which is a noisy measurement of the transcription level X_t at position t in the genome;

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Note: 'functional' definition of state-space models is less convenient in this case.

Stochastic volatility (basic model)

Y_t is log-return of asset price, $Y_t = \log(p_t/p_{t-1})$,

$$Y_t | X_t = x_t \sim \mathcal{N}\left(0, \exp(x_t)\right)$$

where $\{X_t\}$ is an auto-regressive process:

$$X_t - \mu = \phi(X_{t-1} - \mu) + U_t, \quad U_t \sim \mathcal{N}(0, \sigma^2)$$

and $\theta = (\mu, \phi, \sigma^2)$.

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and $\theta = (\mu, \phi, \sigma^2)$.
Take $|\phi| < 1$ and $X_0 \sim N(\mu, \sigma^2/(1-\rho^2))$ to impose stationarity.

Stochastic volatility (variations)

- Student dist' for noises
- skewness: $Y_t = \alpha X_t + \exp(X_t/2) V_t$
- leverage effect: correlation between U_t and V_t
- multivariate extensions

Nonlinear dynamic systems in Ecology, Epidemiology, and other fields

$Y_t = X_t + V_t$, where $\{X_t\}$ is some complex nonlinear dynamic system. In Ecology for instance,

$$X_t = X_{t-1} + \theta_1 - \theta_2 \exp(\theta_3 X_{t-1}) + U_t$$

where X_t is log of population size. For some values of θ, process is nearly chaotic.

Nonlinear dynamic systems: Lokta-Volterra

Predator-prey model, where $\mathcal{X} = (\mathbb{Z}^+)^2$, $X_t(1)$ is the number of preys, $X_t(2)$ is the number of predators, and, working in continuous-time:

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\begin{aligned}
X_t(1) &\xrightarrow{\theta_1} 2X_t(1) \\
X_t(1) + X_t(2) &\xrightarrow{\theta_2} 2X_t(2), \quad t \in \mathbb{R}^+ \\
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(but Y_t still observed in discrete time.)

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Intractable dynamics: We can simulate from $X_t|X_{t-1}$, but we can't compute $p_t(x_t|x_{t-1})$.

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Intractable dynamics: We can simulate from $X_t|X_{t-1}$, but we can't compute $p_t(x_t|x_{t-1})$.
see also compartmental models in Epidemiology.

State-space models with an intractable or degenerate observation process

We have seen models such that $X_t|X_{t-1}$ is intractable; $Y_t|X_t$ may be intractable as well. Let

$$X'_t = (X_t, Y_t), \quad Y'_t = Y_t + V_t, \quad V_t \sim \mathcal{N}(0, \sigma^2)$$

and use $\{(X'_t, Y'_t)\}$ as an approximation of $\{(X_t, Y_t)\}$.

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and use $\{(X'_t, Y'_t)\}$ as an approximation of $\{(X_t, Y_t)\}$.
$\Rightarrow$ Connection with ABC (likelihood-free inference).

Finite state-space models (aka hidden Markov models)

$\mathcal{X} = \{1, \ldots, K\}$, uses in e.g.

- speech processing; X_t is a word, Y_t is an acoustic measurement (possibly the earliest application of HMMs). Note K is quite large.
- time-series modelling to deal with heterogenity (e.g. in medecine, X_t is state of patient)
- rediscovered in Economics as Markov-switching models; there X_t is the state of the Economy (recession, growth), and Y_t is some economic indicator (e.g. GDP) which follows an ARMA process (with parameters that depend on X_t)
- also related: change-point models

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- also related: change-point models

Note: Not of direct interest to us, as sequential analysis may be performed *exactly* using Baum-Petrie filter.

A quick note on the generality of the definition

Consider a GARCH model, i.e. $Y_t \sim \mathcal{N}(0, \sigma_t^2)$, with

$$\sigma_t^2 = \alpha + \beta Y_{t-1}^2 + \gamma \sigma_{t-1}^2.$$

If we replace $\theta = (\alpha, \beta, \gamma)$ by Markov process (θ_t), do we obtain a state-space model?

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Definition

The phrase *state-space models* refers not only to its definition (in terms of $\{X_t\}$ and $\{Y_t\}$) but also to a particular *inferential scenario*: $\{Y_t\}$ is observed (data denoted $y_0, \ldots$), $\{X_t\}$ is not, and one wishes to recover the X_t's given the Y_t's, often sequentially (over time).

Filtering, prediction, smoothing

Conditional distributions of interest (at every time t)

- Filtering: $X_t | Y_{0:t}$
- Prediction: $X_t | Y_{0:t-1}$
- data prediction: $Y_t | Y_{0:t-1}$
- fixed-lag smoothing: $X_{t-h:t} | Y_{0:t}$ for $h \geq 1$
- complete smoothing: $X_{0:t} | Y_{0:t}$
- likelihood factor: density of $Y_t | Y_{0:t-1}$ (so as to compute the full likelihood)

Parameter estimation

All these tasks are usually performed for a fixed θ (assuming the model depends on some parameter θ). To deal additionally with parameter uncertainty, we could adopt a Bayesian approach, and consider e.g. the law of (θ, X_t) given $Y_{0:t}$ (for filtering). But this is often more involved.

Our notations (spoiler!)

- $\{X_t\}$ is a Markov process with initial law $P_0(\mathrm{d}x_0)$, and Markov kernel $P_t(x_{t-1}, \mathrm{d}x_t)$.
- $\{Y_t\}$ has conditional distribution $F_t(x_t, \mathrm{d}y_t)$, which admits probability density $f_t(y_t|x_t)$ (with respect to common dominating measure $\nu(\mathrm{d}y_t)$).
- when needed, dependence on θ will be made explicit as follows: $P_t^\theta(x_{t-1}, \mathrm{d}x_t)$, $f_t^\theta(y_t|x_t)$, etc.

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Algorithms, calculations, etc may be extended straightforwardly to non-standard situations such that $\mathcal{X}$, $\mathcal{Y}$ vary over time, or such that $Y_t|X_t$ also depends on $Y_{0:t-1}$, but for simplicity, we will stick to these notations.

problems with a structure similar to the sequential analysis of a state-space model

Consider the simulation of Markov process $\{X_t\}$, conditional on $X_t \in A_t$ for each t.

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Consider the simulation of Markov process $\{X_t\}$, conditional on $X_t \in A_t$ for each t.
Take $Y_t = \mathbb{1}(X_t \in A_t)$, $y_t = 1$, then this tasks amounts to smoothing the corresponding state-space model.

A particular example: self-avoiding random walk

Consider a random walk in $\mathbb{Z}^2$, (i.e. at each time we may move north, south, east or west, wit probability $1/4$). We would to simulate $\{X_t\}$ conditional on the trajectory $X_{0:T}$ never visiting the same point more than once.

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How to define $\{X_t\}$ in this case?

Markov processes

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Summary

- Introduce Markov processes via kernels
- Recursions of marginal distributions
- Conditional distributions
  - conditional independence
  - partially observed Markov processes & state-space models
- Graphical models

Definition

A probability kernel from $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ to $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$, $P(x, \mathrm{d}y')$, is a function from $(\mathcal{X}, \mathcal{B}(\mathcal{Y}))$ to $[0, 1]$ such that

(a) for every x, $P(x, \cdot)$ is a probability measure on $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$,

(b) for every $A \in \mathcal{B}(\mathcal{Y})$, $P(x, A)$ is a measurable function in $\mathcal{X}$.

Then, if

$$\mathbb{P}_1(\mathrm{d}x_{0:1}) = \mathbb{P}_0(\mathrm{d}x_0) P_1(x_0, \mathrm{d}x_1)$$

by construction, $\mathbb{P}_1(\mathrm{d}x_0) = \mathbb{P}_0(\mathrm{d}x_0)$ and

$$\mathbb{P}_1(X_1 \in \mathrm{d}x_1 | X_0 = x_0) = P_1(x_0, \mathrm{d}x_1)\,.$$

Backward kernel - Bayes

$$\mathbb{P}_1(\mathrm{d}x_0)P_1(x_0,\mathrm{d}x_1) = \mathbb{P}_1(\mathrm{d}x_1)\overleftarrow{P}_0(x_1,\mathrm{d}x_0)\,,$$

Definition

A sequence of random variables $X_{0:T}$ with joint distribution given by

$$\mathbb{P}_T(X_{0:T} \in \mathrm{d}x_{0:T}) = \mathbb{P}_0(\mathrm{d}x_0) \prod_{s=1}^{T} P_s(x_{s-1}, \mathrm{d}x_s)\,,$$

is called a (discrete-time) Markov process with state-space $\mathcal{X}$, initial distribution $\mathbb{P}_0$ and transition kernel at time t, P_t. Likewise, a probability measure decomposed into a product of an initial distribution and transition kernels as in (2) will be called a Markov measure.

Conditional independence

$$\mathbb{P}_T(X_t \in \mathrm{d}x_t | X_{0:t-1} = x_{0:t-1}) = P_t(x_{t-1}, \mathrm{d}x_t)\,.$$

$$\mathbb{P}_T(X_t \in \mathrm{d}x_t | X_{0:s} = x_{0:s}) = P_{s+1:t}(x_s, \mathrm{d}x_t)\,, \quad \forall t \leq T, s < t\,,$$

where

$$P_{s+1:t}(x_s, A) = \int_{\mathcal{X}^{t-s}} P_{s+1}(x_s, \mathrm{d}x_{s+1}) P_{s+2}(x_{s+1}, \mathrm{d}x_{s+2}) \cdots P_t(x_{t-1}, A)\,.$$

A marginalisation property

Proposition

Consider a sequence of probability measures, index by t, defined as:

$$\mathbb{P}_t(X_{0:t} \in \mathrm{d}x_{0:t}) = \mathbb{P}_0(\mathrm{d}x_0) \prod_{s=1}^{t} P_s(x_{s-1}, \mathrm{d}x_s)\,,$$

where P_s are probability kernels. Then, for any $t \leq T$,

$$\mathbb{P}_T(\mathrm{d}x_{0:t}) = \mathbb{P}_t(\mathrm{d}x_{0:t})\,.$$

Some recursions

$$\mathbb{P}_t(X_t \in \mathrm{d}x_t) = \mathbb{E}_{\mathbb{P}_t}[\mathbb{P}_t(X_t \in \mathrm{d}x_t | X_{0:s})] = \mathbb{E}_{\mathbb{P}_t}[P_{s+1:t}(X_s, \mathrm{d}x_t)] \,.$$

With the marginalisation it yields the Chapman-Kolmogorov equation

$$\mathbb{P}_t(X_t \in \mathrm{d}x_t) = \mathbb{E}_{\mathbb{P}_s}[P_{s+1:t}(X_s, \mathrm{d}x_t)] \,, \quad \forall s \leq t-1 \,.$$

Backward process

$$\mathbb{P}_T(X_{0:T} \in \mathrm{d}x_{0:T}) = \mathbb{P}_T(\mathrm{d}x_T) \prod_{s=1}^{T} \overleftarrow{P}_{T-s}(x_{T-s+1}, \mathrm{d}x_{T-s}) \,,$$

POMP & SSM

$$\mathbb{P}_T(X_{0:T} \in \mathrm{d}x_{0:T}, Y_{0:T} \in \mathrm{d}y_{0:T}) = \mathbb{P}_T(\mathrm{d}x_{0:T}) \prod_{t=0}^{T} f_t(y_t|x_t) \prod_{t=0}^{T} \nu(\mathrm{d}y_t)$$
$$= \mathbb{P}_0(\mathrm{d}x_0) \prod_{t=1}^{T} P_t(x_{t-1}, \mathrm{d}x_t) \prod_{t=0}^{T} f_t(y_t|x_t) \prod_{t=0}^{T} \nu(\mathrm{d}y_t)$$

When relevant, $f_t^\theta(y_t|x_t)$ and $P_t^\theta(x_{t-1}, \mathrm{d}x_t)$

Components of a SSM

Likelihood

$$p_t(y_{0:t}) = \mathbb{E}_{\mathbb{P}_t}\left[\prod_{s=0}^{t} f_s(y_s|x_s)\right]$$

is the density (likelihood/partition function) of the law of $Y_{0:T}$;
Likelihood factors

$$p_t(y_{0:t}) = p_0(y_0)\prod_{s=1}^{t} p_s(y_s|y_{0:s-1})\,.$$

and

$$p_{t+k}(y_{t:t+k}|y_{0:t-1}) = p_{t+k}(y_{0:t+k})/p_{t-1}(y_{0:t-1})\,, \quad k \geq 0\,,\ t \geq 1\,.$$

Law of states given observations

$$\mathbb{P}_t(X_{0:t} \in \mathrm{d}x_{0:t} | Y_{0:t} = y_{0:t}) = \frac{1}{p_t(y_{0:t})} \left\{ \prod_{s=0}^{t} f_s(y_s|x_s) \right\} \mathbb{P}_t(\mathrm{d}x_{0:t}) \,.$$

(To see this, multiply both sides by $p_t(y_{0:t}) \prod_{s=0}^{t} \nu(\mathrm{d}y_s)$)
Another SSM function that will be is likelihood of future observations given the current value of the state.

$$p_T(y_{t+1:T}|x_t) = \frac{\mathbb{P}_T(Y_{t+1:T} \in \mathrm{d}y_{t+1:T} | X_{0:t} = x_{0:t}, Y_t = y_t)}{\nu^{T-t}(\mathrm{d}y_{t+1:T})} \,, t < T \,,$$

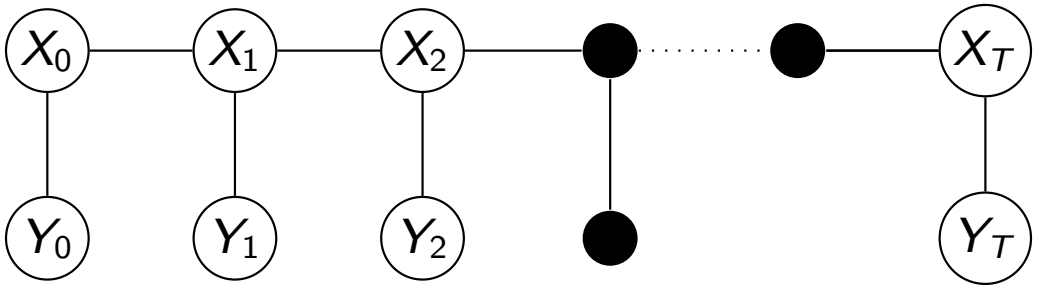
where by conditional independence it does not depend on $x_{0:t-1}, y_t$

Restating SSM aims

- **state prediction**: deriving $\mathbb{P}_t(X_{t+1:t+h} \in \mathrm{d}x_{t+1:t+h} | Y_{0:t} = y_{0:t})$, for $h \geq 1$;
- **filtering**: deriving $\mathbb{P}_t(X_t \in \mathrm{d}x_t | Y_{0:t} = y_{0:t})$;
- **fixed-lag smoothing**: deriving $\mathbb{P}_t(X_{t-l:t} \in \mathrm{d}x_{t-l:t} | Y_{0:t} = y_{0:t})$ for some $l \geq 1$;
- **(complete) smoothing**: deriving $\mathbb{P}_t(X_{0:t} \in \mathrm{d}x_{0:t} | Y_{0:t} = y_{0:t})$;
- **likelihood computation**: deriving $p_t(y_{0:t})$.

Graphical models

Variables as nodes; when any two are linked by a kernel draw an edge:



Path; conditional independence; Markov property of $(X_{0;T}, Y_{0:T})$, $X_{0:T}$, $X_{0:T}$ conditionally on $Y_{0:T}$ but not of $Y_{0:T}$.

Further reading

- Conditional independence, Chapter 5 of *Foundations of modern Probability* (Kallenberg, Springer)
- Intro to graphical models: Chapter 8 of *Pattern recognition and machine learning* (Bishop, Springer)

Feynman-Kac models & HMMs

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Outline

1. Feynman-Kac models
   - Change of measure
   - Feynman-Kac formalism
   - Feynman-Kac formalisms of a state space model
   - Forward recursion
   - FK as Markov measures
2. HMMs

Summary

- Tool: change of measure
- Define FK models via Markov and CoM
- FK formalism of given probabilistic models
- Explore properties of FK models: recursion, marginalisation, Markovianity
- Apply the machinery on specific SSMs: HMMs.

Change of measure

Definition

Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable space, and $\mathbb{M}$ and $\mathbb{Q}$ two probability measures defined on this space. We then say that $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{M}$, if for any $A \in \mathcal{B}(\mathcal{X})$ for which $\mathbb{M}(A) = 0$, $\mathbb{Q}(A) = 0$. In this case, we also say that $\mathbb{M}$ dominates $\mathbb{Q}$.

In fact, $\mathbb{Q}$ is a.c. wrt to $\mathbb{M}$ iff $\exists$

$$w(x) = \frac{\mathbb{Q}(\mathrm{d}x)}{\mathbb{M}(\mathrm{d}x)}$$

(Radon-Nikodym)

Suppose that $\mathbb{Q}$ and $\mathbb{M}$ are probability mesures on a space $\mathcal{X}$, and $w(x) \propto \mathbb{Q}(\mathrm{d}x)/\mathbb{M}(\mathrm{d}x)$. Then, for any test function ϕ,

$$\mathbb{M}(\phi w) = \mathbb{Q}(\phi)\mathbb{M}(w)\,.$$

The other way around

Alternatively, if you give me:

- A probability measure $\mathbb{M}$;
- a function $G \geq 0$ such that

$$L := \mathbb{M}(G) \in (0, \infty)$$

Then I can define:

$$\mathbb{Q}(\mathrm{d}x) = \frac{1}{L}\mathbb{M}(\mathrm{d}x)G(x)$$

The components of a Feynman-Kac model

- Markov measure:

$$\mathbb{M}_T(\mathrm{d}x_{0:T}) = \mathbb{M}_0(\mathrm{d}x_0) \prod_{t=1}^{T} M_t(x_{t-1}, \mathrm{d}x_t) \,.$$

- Potential functions, $G_0 : \mathcal{X} \to \mathbb{R}^+$, and $G_t : \mathcal{X}^2 \to \mathbb{R}^+$, for $1 \leq t \leq T$
- Change of measure: for $t \leq T$

$$\mathbb{Q}_t(\mathrm{d}x_{0:T}) = \frac{1}{L_t} G_0(x_0) \left\{ \prod_{s=1}^{t} G_s(x_{s-1}, x_s) \right\} \mathbb{M}_T(\mathrm{d}x_{0:T})$$

Components: $T, G_0, \mathbb{M}_0,\ G_t(x_{t-1}, x_t), M_t(x_{t-1}, \mathrm{d}x_t)$

Partition function/evidence/marginal likelihood

$$L_t = \int_{\mathcal{X}^{t+1}} G_0(x_0) \prod_{s=1}^{t} G_s(x_{s-1}, x_s) \, \mathbb{M}_t(\mathrm{d}x_{0:t})$$
$$= \mathbb{E}_{\mathbb{M}_t} \left[G_0(X_0) \prod_{s=1}^{t} G_s(X_{s-1}, X_s) \right] .$$

and assume that G_t's such that $0 < L_t < \infty$ for all t
Normalising factors: $\ell_t = L_t / L_{t-1}$

The "bootstrap" Feynman-Kac formalism of a state-space model

For a state-space model with transition kernels $P_t(x_{t-1}, \mathrm{d}x_t)$ observation densities $f_t(y_t|x_t)$, define its "bootstrap" Feynman-Kac formalism as the Feynman-Kac model such that

$$M_t(x_{t-1}, \mathrm{d}x_t) = P_t(x_{t-1}, \mathrm{d}x_t)\,, \quad G_t(x_{t-1}, x_t) = f_t(y_t|x_t)\,,$$

and $\mathbb{M}_0(\mathrm{d}x_0) = \mathbb{P}_0(\mathrm{d}x_0)$. Then:

$$\begin{aligned}
\mathbb{Q}_{t-1}(\mathrm{d}x_{0:t}) &= \mathbb{P}_t(X_{0:t} \in \mathrm{d}x_{0:t} | Y_{0:t-1} = y_{0:t-1}) \\
\mathbb{Q}_t(\mathrm{d}x_{0:t}) &= \mathbb{P}_t(X_{0:t} \in \mathrm{d}x_{0:t} | Y_{0:t} = y_{0:t}) \\
L_t &= p_t(y_{0:t})
\end{aligned}$$

Is this the only one? And what is this formalism useful for?

The "guided" Feynman-Kac formalism of a state-space model

Consider a state-space model with signal transition kernels $P_t(x_{t-1}, \mathrm{d}x_t)$ and observation densities $f_t(y_t|x_t)$. We define its "guided" Feynman-Kac formalism to be a Feynman-Kac model with the following components

$$G_0(x_0)\mathbb{M}_0(\mathrm{d}x_0) = f_0(y_0|x_0)\mathbb{P}_0(\mathrm{d}x_0)\,,$$
$$G_t(x_{t-1}, x_t)M_t(x_{t-1}, \mathrm{d}x_t) = f_t(y_t|x_t)P_t(x_{t-1}, \mathrm{d}x_t)\,.$$

meaning of equalities, special case

The "auxiliary" Feynman-Kac formalism of a state-space model

Consider a state-space model with signal transition kernels $P_t(x_{t-1}, \mathrm{d}x_t)$ and observation densities $f_t(y_t|x_t)$. Additionally, let $\eta_t(x_t)$ be user-chosen, "auxiliary" functions, such that $\mathbb{E}_{\mathbb{P}_t}[\eta_t(X_t)|Y_{0:t} = y_{0:t}] < \infty$ for all t. We define its "auxiliary" Feynman-Kac formalism to be a Feynman-Kac model with the following components

$$G_0(x_0)\mathbb{M}_0(\mathrm{d}x_0) = f_0(y_0|x_0)\mathbb{P}_0(\mathrm{d}x_0)\eta_0(x_0)$$

$$G_t(x_{t-1}, x_t)M_t(x_{t-1}, \mathrm{d}x_t) = f_t(y_t|x_t)P_t(x_{t-1}, \mathrm{d}x_t)\frac{\eta_t(x_t)}{\eta_{t-1}(x_{t-1})}$$

Use of formalism

- Decouple a statistical model (the state-space model) from its mathematical representation $\rightarrow$ unified treatment of theory (recursions) and numerics (particle filters)
- Feynman-Kac models share the same fundamental structure: the specific change of measure from a Markov measure $\rightarrow$ common set of recursions regardless of the details of components
- Feynman-Kac representation and modularity
- Feynman-Kac outside state-space models

Forward recursion (Feynman-Kac formalism) pt1

Initialise with $\mathbb{Q}_{-1}(\mathrm{d}x_0) = \mathbb{M}_0(\mathrm{d}x_0)$, then, for $t = 0 : T$,

- Extension:

$$\mathbb{Q}_{t-1}(\mathrm{d}x_{t-1:t}) = \mathbb{Q}_{t-1}(\mathrm{d}x_{t-1}) M_t(x_{t-1}, \mathrm{d}x_t)$$

  Recall ▸ definition

- Change of measure:

$$\mathbb{Q}_t(\mathrm{d}x_{t-1:t}) = \frac{1}{\ell_t} G_t(x_{t-1}, x_t) \mathbb{Q}_{t-1}(\mathrm{d}x_{t-1:t})$$

Forward recursion (Feynman-Kac formalism) pt2

with

$$\ell_0 = L_0 = \int_{\mathcal{X}} G_0(x_0) M_0(\mathrm{d}x_0)$$

and

$$\ell_t = \frac{L_t}{L_{t-1}} = \int_{\mathcal{X}^2} G_t(x_{t-1}, x_t) \mathbb{Q}_{t-1}(\mathrm{d}x_{t-1:t})$$

for $t \geq 1$.

- Marginalisation:

$$\begin{aligned} \mathbb{Q}_t(\mathrm{d}x_t) &= \int_{\mathcal{X}} \mathbb{Q}_t(\mathrm{d}x_{t-1:t}) \\ &= \frac{1}{\ell_t} \int_{\mathcal{X}} G_t(x_{t-1}, x_t) M_t(x_{t-1}, \mathrm{d}x_t) \mathbb{Q}_{t-1}(\mathrm{d}x_{t-1}) \end{aligned}$$

Implications for the "b"-fm: recursion for filter, prediction, likelihood

$$\mathbb{P}_{t-1}(X_t \in \mathrm{d}x_t | Y_{0:t-1} = y_{0:t-1}) = \int_{\mathcal{X}} P_t(x_{t-1}, \mathrm{d}x_t) \mathbb{P}_t(X_{t-1} \in \mathrm{d}x_{t-1} | Y_{0:t-1} = y_{0:t-1}),$$

$$\mathbb{P}_t(X_t \in \mathrm{d}x_t | Y_{0:t} = y_{0:t}) = \frac{1}{p_t(y_t | y_{0:t-1})} f_t(y_t | x_t) \mathbb{P}_{t-1}(X_t \in \mathrm{d}x_t | Y_{0:t-1} = y_{0:t-1}).$$

$$p_t(y_t | y_{0:t-1}) = \int_{\mathcal{X}^2} f_t(y_t | x_t) \mathbb{P}_{t-1}(X_{t-1:t} \in \mathrm{d}x_{t-1:t} | Y_{0:t-1} = y_{0:t-1}).$$

Feynman-Kac model as a Markov measure - cost-to-go functions

$$H_{T:T}(x_T) = 1\,,$$

$$H_{t:T}(x_t) = \int_{\mathcal{X}^{T-t}} \prod_{s=t+1}^{T} G_s(x_{s-1}, x_s) M_s(x_{s-1}, \mathrm{d}x_s)\,, \quad t < T\,.$$

Hence

$$H_{t:T}(x_t) = \int_{\mathcal{X}} G_{t+1}(x_t, x_{t+1}) H_{t+1:T}(x_{t+1}) M_{t+1}(x_t, \mathrm{d}x_{t+1})$$

but also

$$H_{t:T}(x_t) = \mathbb{E}_{\mathbb{M}_T}\left[\prod_{s=t+1}^{T} G_s(X_{s-1}, X_s) \middle| X_t = x_t\right]$$

$$= \mathbb{E}_{\mathbb{M}_{t+1}}\left[G_{t+1}(X_t, X_{t+1}) H_{t+1:T}(X_{t+1}) | X_t = x_t\right].$$

Proposition

$\mathbb{Q}_T$ is the law of a Markov process with state-space $\mathcal{X}$, initial distribution

$$\mathbb{Q}_{0|T}(\mathrm{d}x_0) = \frac{H_{0:T}(x_0)}{L_T} G_0(x_0)\mathbb{M}_0(\mathrm{d}x_0)\,,$$

forward transition kernels $Q_{t|T}(x_{t-1}, \mathrm{d}x_t)$ given by:

$$Q_{t|T}(x_{t-1}, \mathrm{d}x_t) = \frac{H_{t:T}(x_t)}{H_{t-1:T}(x_{t-1})} G_t(x_{t-1}, x_t) M_t(x_{t-1}, \mathrm{d}x_t)\,,$$

and backward kernels given by:

$$\overleftarrow{Q}_{t-1}(x_t, \mathrm{d}x_{t-1}) = \frac{Q_{t|T}(x_{t-1}, \mathrm{d}x_t)}{\mathbb{Q}_T(\mathrm{d}x_t)} \mathbb{Q}_T(\mathrm{d}x_{t-1})\,.$$

Implications for the "b"-fm: POMP

By immediate translation:

$$H_{t:T}(x_t) = \frac{\mathbb{P}_T(Y_{t+1:T} \in \mathrm{d}y_{t+1:T} | X_t = x_t)}{\nu^{T-t}(\mathrm{d}y_{t+1:T})}, t < T.$$

$$p_t(y_{t+1:T}|x_t) = \int_{\mathcal{X}} f(y_{t+1}|x_{t+1}) p(y_{t+2:T}|x_{t+1}) P_t(x_t, \mathrm{d}x_{t+1}).$$

Hence, the conditioned Markov process is also Markov with

$$\mathbb{P}_{0|T}(\mathrm{d}x_0) = \frac{p(y_{1:T}|x_0)}{p(y_{0:T})} f_0(y_0|x_0) \mathbb{P}_0(\mathrm{d}x_0),$$

$$P_{t|T}(x_{t-1}, \mathrm{d}x_t) = \frac{p(y_{t+1:T}|x_t)}{p(y_{t:T}|x_{t-1})} f_t(y_t|x_t) P_t(x_{t-1}, \mathrm{d}x_t).$$

Stability properties

Forward-backward recursions in Feynman-Kac models

Recall that $\mathbb{M}_T(\mathrm{d}x_{0:t}) = \mathbb{M}_t(\mathrm{d}x_{0:t})$. For the Feynman-Kac model we have:

Proposition

For any $t < T$,

$$\mathbb{Q}_T(\mathrm{d}x_{0:t}) = \frac{L_t}{L_T} H_{t:T}(x_t) \mathbb{Q}_t(\mathrm{d}x_{0:t}) \,.$$

Ideas for proof?

Proof: use the Markov property of the Q process

Corollary

$$\mathbb{Q}_T(\mathrm{d}x_t) = \frac{L_t}{L_T} H_{t:T}(x_t)\mathbb{Q}_t(\mathrm{d}x_t)\,.$$

& from Proposition 2 and the result above we get:

Corollary

$$\overleftarrow{Q}_{t-1}(x_t, \mathrm{d}x_{t-1}) = \frac{1}{\ell_t} G_t(x_{t-1}, x_t) \frac{M_t(x_{t-1}, \mathrm{d}x_t)}{\mathbb{Q}_t(\mathrm{d}x_t)} \mathbb{Q}_{t-1}(\mathrm{d}x_{t-1})$$

Implications for the "b"-fm: forward filtering/backward smoothing

$$\mathbb{P}(X_t \in \mathrm{d}x_t | Y_{0:T} = y_{0:T}) = \frac{1}{p(y_{t+1:T}|y_{0:t})} p(y_{t+1:T}|x_t) \mathbb{P}(X_t \in \mathrm{d}x_t | Y_{0:t} = y_{0:t})$$

$$\overleftarrow{P}_{t|t}(x_{t+1}, \mathrm{d}x_t) = \frac{1}{p(y_t|y_{0:t-1})} f_t(y_t|x_t) \frac{P_t(x_{t-1}, \mathrm{d}x_t)}{\mathbb{P}_t(X_t \in \mathrm{d}x_t | Y_{0:t} = y_{0:t})} \mathbb{P}_{t-1}(X_{t-1} \in \mathrm{d}x_{t-1} | Y_{0:t-1} = y_{0:t-1}) .$$

Forward-backward simulation

How generate draws from $\mathbb{Q}_T(\mathrm{d}x_{0:T})$?

Then, we should know how to generate from

$$\mathbb{P}_T(X_{0:T} \in \mathrm{d}x_{0:T} | Y_{0:T} = y_{0:T})$$

ideas?

Further reading

- *Feynman-Kac formulae* (Del Moral, Springer)

Outline

1 Feynman-Kac models
- Change of measure
- Feynman-Kac formalism
- Feynman-Kac formalisms of a state space model
- Forward recursion
- FK as Markov measures

2 HMMs

HMMs

- $\mathcal{X} = \{1, \ldots, K\}$
- Integrals $\rightarrow$ sums; measures $\rightarrow$ vectors; kernels $\rightarrow$ matrices
- Following based on "bootstrap" formalism

Recursions for hidden Markov models

The following are understood for all $k \in 1 : \mathcal{D}$

- **Prediction:**

$$\mathbb{P}_{t-1}(X_t = k | Y_{0:t-1} = y_{0:t-1}) = \sum_{l=1}^{\mathcal{D}} \mathbb{P}_{t-1}(X_{t-1} = l | Y_{0:t-1} = y_{0:t-1}) p_t(k|l)$$

- **Filter:**

$$\mathbb{P}_t(X_t = k | Y_{0:t} = y_{0:t}) = \frac{1}{p(y_t | y_{0:t-1})} \mathbb{P}_{t-1}(X_t = k | Y_{0:t-1} = y_{0:t-1}) f_t(y_t|k)$$

- **Likelihood factors:**

$$p_t(y_t | y_{0:t-1}) = \sum_k \mathbb{P}_{t-1}(X_t = k | Y_{0:t-1} = y_{0:t-1}) f_t(y_t|k)$$

- **Likelihood of future observations given current state:**

$$p_T(y_{t+1:T}|k) = \sum_l f(y_{t+1}|l) p(y_{t+2:T}|l) p_t(l|k)$$

Complexity

- Predictive probabilities: $\mathcal{O}(K^2)$ (unless sparse transition matrix, e.g. change point models)
- Given those, filter & likelihood factors obtained at $\mathcal{O}(K)$
- Overall cost: $\mathcal{O}(TK^2)$ as opposed to $\mathcal{O}(K^T)$
- Still, K might be large...

chapter chapter

Intro to IS & resampling

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(based on a previous PG course with O. Papaspiliopoulos)

Summary

- Introduce MC integration & basic perspective on IS (incl. some optimality considerations)
- Shift attention to probability measure approximations
- Study MSE of IS: asymptotics and concentration inequalities
- Measures of performance: ESS
- Random weight importance sampling: a forest of algorithms!
- CoD and introduction to the tool of resampling

Outline

1. Intro to MC
2. IS
   - A first take
   - MSE & ESS
   - Random Weight IS
3. Introduction to resampling
   - Motivating examples
   - Resampling framework

101 of MC

$$\mathbb{E}_q[\varphi(X)] = \int_{\mathcal{X}} \varphi(x) q(x) \, \mathrm{d}x \quad \approx \frac{1}{N} \sum_{n=1}^{N} \varphi(X^n), \qquad X^n \sim q,$$

Assessing the error: if $\mathbb{E}_q[\varphi(X)^2] < +\infty$

$$\mathrm{MSE}\left[\frac{1}{N}\sum_{n=1}^{N}\varphi(X^n)\right] = \mathbb{E}\left[\left\{\frac{1}{N}\sum_{n=1}^{N}\varphi(X^n) - \mathbb{E}_q[\varphi(X)]\right\}^2\right] = \frac{1}{N}\mathrm{Var}_q[\varphi(X)]$$

1. error vanishes at rate $O_P(N^{-1/2})$, which is rather slow: to divide the error by 10, you need 100 more samples;
2. O_P means that the bound on the approximation error is stochastic: error is below a threshold with some probability

$$LLN \quad \left\{ \frac{1}{N} \sum_{n=1}^{N} \varphi(X^n) - \mathbb{E}_q[\varphi(X)] \right\} \to 0 \quad \text{a.s.}$$

$$CLT \quad N^{1/2} \left\{ \frac{1}{N} \sum_{n=1}^{N} \varphi(X^n) - \mathbb{E}_q[\varphi(X)] \right\} \Rightarrow \mathcal{N}\left(0, \mathrm{Var}_q[\varphi(X)]\right),$$

$$CI \quad \left[\frac{1}{N} \sum_{n=1}^{N} \varphi(X^n) \pm \frac{z_{1-\alpha/2}}{N^{1/2}} \mathrm{Var}_q[\varphi(X)] \right]$$

Issues:

- Plug in estimators
- Concentration inequalities
- What if $\mathbb{E}_q[\varphi(X)^2] = \infty$
- Math Finance/Physics vs Stats

Outline

1 Intro to MC

2 IS
- A first take
- MSE & ESS
- Random Weight IS

3 Introduction to resampling
- Motivating examples
- Resampling framework

A first take on IS

$$\int_{\mathcal{X}} \varphi(x) q(x) \mathrm{d}x = \int_{\mathcal{X}} \varphi(x) \frac{q(x)}{m(x)} m(x) \mathrm{d}x$$

Sufficient condition for validity

- Normalised IS estimator

$$\frac{1}{N} \sum_{n=1}^{N} w(X^n) \varphi(X^n) \quad X^n \sim m \quad w(x) = q(x)/m(x)$$

Alternatively,

$$\int_{\mathcal{X}} \varphi(x) q(x) \, \mathrm{d}x = \frac{\int_{\mathcal{X}} \varphi(x) \frac{q(x)}{m(x)} m(x) \, \mathrm{d}x}{\int_{\mathcal{X}} \frac{q(x)}{m(x)} m(x) \, \mathrm{d}x} = \frac{\int_{\mathcal{X}} \varphi(x) \frac{q_u(x)}{m_u(x)} m(x) \, \mathrm{d}x}{\int_{\mathcal{X}} \frac{q_u(x)}{m_u(x)} m(x) \, \mathrm{d}x}$$

- Autonormalised IS estimator

$$\sum_{n=1}^{N} W^n \varphi(X^n), \qquad W^n = \frac{w(X^n)}{\sum_{m=1}^{N} w(X^m)}.$$

Question

Which one??

Optimal proposal

Proposition

The MSE of the normalised IS estimator of $\mathbb{E}_q[\varphi(X)]$, is minimised with respect to m by taking $m = m^\star$,

$$m^\star(x) = \frac{q(x)\,|\varphi(x)|}{\int_{\mathcal{X}} |\varphi(x')|\, q(x')\,\mathrm{d}x'}.$$

Proof:

$$\begin{aligned}
\mathrm{Var}_m\left[q(X)\varphi(X)/m(X)\right] &= \mathbb{E}_m\left[q(X)^2\varphi(X)^2/m(X)^2\right] - \mathbb{E}_q[\varphi(X)]^2 \\
&\geq \left\{\mathbb{E}_q\left[|\varphi(X)|\right]\right\}^2 - \mathbb{E}_q[\varphi(X)]^2 \\
&= \mathrm{Var}_{m^\star}\left[q(X)\varphi(X)/m^\star(X)\right].
\end{aligned}$$

- If $\varphi \geq 0$ the variance of the optimal normalised estimator is 0
- Generally

$$w^\star(x)\varphi(x) = \frac{q(x)}{m^\star(x)}\varphi(x) = \mathrm{sign}(\phi(x))\mathbb{E}_q[|\varphi(X)|]$$

- Signed test functions 0 variance
- Auto-normalised:

$$m^\star(x) = \frac{q(x)|\overline{\varphi}(x)|}{\int_{\mathcal{X}} q(x')|\overline{\varphi}(x')|\,\mathrm{d}x'}$$

- Particle approximation vs tailored estimation

Formal perspective on IS

1. $\mathbb{Q}$ vs q
2. CoM:

$$\mathbb{M}(\phi w) = \mathbb{Q}(\phi)\mathbb{M}(w)\,.$$

that suggests

$$\sum_{n=1}^{N} W^n \varphi(X^n), \qquad W^n = \frac{w(X^n)}{\sum_{m=1}^{N} w(X^m)}\,.$$

3. Estimator vs particle approximation; random probability measure

$$\mathbb{Q}^N(\mathrm{d}x) := \sum_{n=1}^{N} W^n \delta_{X^n}(\mathrm{d}x), \qquad X^n \sim \mathbb{M}\,.$$

Dirac

Then

$$\mathbb{Q}^N(\varphi) = \sum_{n=1}^{N} W^n \varphi(X^n)\,.$$

and we can think of *weak convergence of measures*, i.e. we say that

$$\mathbb{Q}^N \Rightarrow \mathbb{Q}$$

as $N \to +\infty$ provided that $\mathbb{Q}^N(\varphi) \to \mathbb{Q}(\varphi)$ for any $\varphi \in \mathcal{C}_b(\mathcal{X})$

MSE of normalised IS

Context: target $\mathbb{Q}$, proposal $\mathbb{M}$, unnormalised weight $w(x)$ and normalised weight $w(x)/\mathbb{M}(w)$.

$$\mathrm{MSE}\left[\frac{1}{N}\sum_{n=1}^{N}\frac{w(X^n)}{\mathbb{M}(w)}\varphi(X^n)\right] = \frac{1}{N}\times\frac{\mathbb{M}[w^2\varphi^2]-\mathbb{Q}(\varphi)^2}{\mathbb{M}(w)^2}\,.$$

For auto-normalised, notice that

$$\mathbb{Q}^N(\varphi)-\mathbb{Q}(\varphi) = \frac{\frac{1}{N}\sum_{n=1}^{N}\frac{w(X^n)}{\mathbb{M}(w)}[\varphi(X^n)-\mathbb{Q}(\varphi)]}{\frac{1}{N}\sum_{n=1}^{N}\frac{w(X^n)}{\mathbb{M}(w)}}$$

Asymptotics

Ratio of RVs:

- nominator: normalised importance sampling estimator of $\mathbb{Q}(\overline{\varphi})$ with $\overline{\varphi} = \varphi - \mathbb{Q}(\varphi)$
- denominator: normalised importance sampling estimator of 1.

Provided

$$\mathbb{M}(w^2\varphi^2) < \infty\,, \tag{1}$$

by CLT

$$\sqrt{N}\left\{\frac{1}{N}\sum_{n=1}^{N}\frac{w(X^n)}{\mathbb{M}(w)}[\varphi(X^n) - \mathbb{Q}(\varphi)]\right\} \Rightarrow \mathcal{N}\left(0, \frac{\mathbb{M}(w^2\overline{\varphi}^2)}{\mathbb{M}(w)^2}\right)\,.$$

and by SLLN

$$\frac{1}{N}\sum_{n=1}^{N}\frac{w(X^n)}{\mathbb{M}(w)} \to 1 \quad \text{a.s.}$$

Hence, by *Slutsky theorems*

$$\sqrt{N}\left\{\mathbb{Q}^N(\varphi) - \mathbb{Q}(\varphi)\right\} \Rightarrow \mathcal{N}\left(0, \frac{\mathbb{M}(w^2\overline{\varphi}^2)}{\mathbb{M}(w)^2}\right)$$

from which we get that for large N,

$$\mathrm{MSE}\left\{\mathbb{Q}^N(\varphi)\right\} \approx \frac{1}{N}\frac{\mathbb{M}(w^2\overline{\varphi}^2)}{\mathbb{M}(w)^2}\,.$$

Slutsky: what is it about?

Note similarity to the MSE of the normalised estimator

Necessary condition for finite variance

Proposition

The condition $\mathbb{M}(w^2\overline{\varphi}^2) < +\infty$, *with* $\overline{\varphi} = \varphi - \mathbb{Q}(\varphi)$, *holds for all bounded functions* φ *if and only if* $\mathbb{M}(w^2) < \infty$.

Variance of the weight good criterion for the performance of IS:

$$\mathrm{MSE}\left\{\mathbb{Q}^N(\varphi)\right\} < \infty \text{ for all bounded functions } \varphi \iff \mathbb{M}(w^2) < \infty\,.$$

Proof.

Sufficient condition: assume $\mathbb{M}(w^2) < \infty$, and take φ such that $|\varphi| < C$. Then $|\overline{\varphi}| \leq 2C$ and $\mathbb{M}(w^2\overline{\varphi}^2) \leq 4C^2\mathbb{M}(w^2) < +\infty$.
Necessary condition: if $\mathbb{Q}$ is a trivial probability measure such that $\mathbb{Q}(A)$ is either 0 or 1 for $A \in \mathcal{B}(\mathcal{X})$, then by absolute continuity so is $\mathbb{M}$ and $\mathbb{M}(w^2) = 1$ and $\mathbb{M}(w^2\overline{\varphi}^2) < +\infty$. Away from this trivial case, we can choose some set $A \in \mathcal{B}(\mathcal{X})$ such that $0 < \mathbb{Q}(A) \leq 1/2$. Then, taking $\varphi = \mathbb{1}_A$, note that

$$\mathbb{M}(w^2\overline{\varphi}^2) = \mathbb{M}(w^2\mathbb{1}_A)(1 - 2\mathbb{Q}(A)) + \mathbb{Q}(A)^2\mathbb{M}(w^2)\,.$$

Therefore, if $\mathbb{M}(w^2) = \infty$ then $\mathbb{M}(w^2\overline{\varphi}^2) = \infty$ for some bounded function φ. $\square$

Non-asymptotic results

A recent article[1] provides the following concentration inequality with strong assumptions on the test but very weak on the weights:

Theorem

$$\sup_{|\varphi|\leq 1} \mathrm{MSE}\{\mathbb{Q}^N(\varphi)\} \leq \frac{4}{N}\frac{\mathbb{M}(w^2)}{\mathbb{M}(w)^2}\,.$$

Directly links with the metric introduced in [2]

not conservative; key quantity

[1] Agapiou, S., Papaspiliopoulos, O., Sanz-Alonso, D., and Stuart, A. (2015). Aspects of importance sampling in high dimensions and low noise regimes. *Unpublished manuscript*

[2] Rebeschini, P. and van Handel, R. (2013). Can local particle filters beat the curse of dimensionality? *arXiv preprint arXiv:1301.6585*

Proof

$$
\begin{aligned}
\mathbb{Q}^N(\varphi) - \mathbb{Q}(\varphi) &= \frac{\frac{1}{N}\sum_n \varphi(X^n)w(X^n)}{\frac{1}{N}\sum_n w(X^n)} - \frac{\mathbb{M}(\varphi w)}{\mathbb{M}(w)} \\
&= \left(\frac{1}{\frac{1}{N}\sum_n w(X^n)} - \frac{1}{\mathbb{M}(w)}\right)\frac{1}{N}\sum_n \varphi(X^n)w(X^n) \\
&- \frac{1}{\mathbb{M}(w)}\left(\mathbb{M}(\varphi w) - \frac{1}{N}\sum_n \varphi(X^n)w(X^n)\right) \\
&= \frac{1}{\mathbb{M}(w)}\left(\mathbb{M}(w) - \frac{1}{N}\sum_n w(X^n)\right)\mathbb{Q}^N(\varphi) \\
&- \frac{1}{\mathbb{M}(w)}\left(\mathbb{M}(\varphi w) - \frac{1}{N}\sum_n \varphi(X^n)w(X^n)\right)
\end{aligned}
$$

Proof ctd

- inequality $(a+b)^2 \leq 2(a^2+b^2)$
- for $|\varphi| \leq 1$, $|\mathbb{Q}^N(\varphi)| \leq 1$ a.s.

to get:

$$\begin{aligned}
\mathrm{MSE}\{\mathbb{Q}^N(\varphi)\} &\leq \frac{2}{\mathbb{M}(w)^2}\left\{\mathrm{MSE}\left\{\frac{1}{N}\sum_n w(X^n)\right\}\right. \\
&\quad \left. + \mathrm{MSE}\left\{\frac{1}{N}\sum_n \varphi(X^n)w(X^n)\right\}\right\} \\
&\leq \frac{2}{N\mathbb{M}(w)^2}\{\mathbb{M}(w^2)+\mathbb{M}(w^2\varphi^2)\} \\
&\leq \frac{4}{N}\frac{\mathbb{M}(w^2)}{\mathbb{M}(w)^2}
\end{aligned}$$

The same article gives a dual, on almost weakest possible on test, but strong on weights:

Theorem

For test functions φ and importance sampling weights w with the obvious sufficient regularity as determined by the right hand side, we have the following bound:

$$\mathrm{MSE}\{\mathbb{Q}^N(\varphi)\} \leq \left(\frac{2}{\mathbb{M}(w)^2} \tilde{C}_2 \mathbb{M}(\varphi^2 w^2) + \frac{4}{\mathbb{M}(w)^4} \mathbb{M}(|\varphi w|^{2d})^{\frac{1}{d}} \tilde{C}_{2e}^{\frac{1}{e}} \mathbb{M}(w^{2e})^{\frac{1}{e}} \right.$$
$$\left. + \frac{4}{\mathbb{M}(w)^{2(1+\frac{1}{p})}} \mathbb{M}(|\varphi|^{2p})^{\frac{1}{p}} \tilde{C}_{2q(1+\frac{1}{p})}^{\frac{1}{q}} \mathbb{M}(w^{2q(1+\frac{1}{p})})^{\frac{1}{q}} \right) \cdot N^{-1}$$

The constants $\tilde{C}_t$, $t \geq 2$ are determined by the Marcinkiewicz-Zygmund inequality, as in Ren and Liang (2001). Furthermore, the pairs of parameters d, e, and p, q are conjugate indices.

ESS

$$\mathrm{ESS}(W^{1:N}) := \frac{1}{\sum_{n=1}^N (W^n)^2} = \frac{\{\sum_{n=1}^N w(X^n)\}^2}{\sum_{n=1}^N w(X^n)^2}$$

$$\frac{N}{\mathrm{ESS}(W^{1:N})} = \frac{N^{-1}\sum_{n=1}^N w(X^n)^2}{\{N^{-1}\sum_{n=1}^N w(X^n)\}^2}$$
$$= 1 + \frac{N^{-1}\sum_{n=1}^N w(X^n)^2 - \{N^{-1}\sum_{n=1}^N w(X^n)\}^2}{\{N^{-1}\sum_{n=1}^N w(X^n)\}^2}.$$

- MC est of MSE; set N for desirable accuracy
- One plus 'coefficient of variation'
- As sample size: $\mathrm{ESS}(W^{1:N}) \in [1, N]$; if k weights equal one, and $N-k$ weights equal zero, then $\mathrm{ESS}(W^{1:N}) = k$; semi(pseudo)-formal justification in Section 2.5.3 of Liu and Chen (2008), "rule of thumb"

ESS and information

- 2nd term converges to chi-square pseudo-distance between $\mathbb{M}$ and $\mathbb{Q}$:

$$\mathbb{M}\left[\left\{\frac{d\mathbb{Q}}{\mathrm{d}\mathbb{M}}-1\right\}^2\right]=\mathrm{Var}_{\mathbb{M}}\left(\frac{w}{\mathbb{M}(w)}\right)$$

- Entropy:

$$\mathrm{Ent}(W^{1:N}):=-\sum_{n=1}^{N}W^n\log(W^n)\,,$$

or instead

$$\sum_{n=1}^{N}W^n\log(NW^n)=\log N-\mathrm{Ent}(W^{1:N})$$

$$\to D_{KL}(\mathbb{Q}||\mathbb{M}):=\int\log\left(\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{M}}\right)\mathrm{d}\mathbb{Q}$$

Random weight methodology

$\mathbb{Q}$ target, $\mathbb{M}$ proposal, $w(x) \propto \mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{M}$. Then,

$$(X^n, \mathcal{W}^n)\,, \quad X^n \sim \mathbb{M}\,, \quad \mathbb{E}[\mathcal{W}^n | X^n] = c\frac{w(X^n)}{\mathbb{M}(w)}\,,$$

for auto-normalised IS; $W^n = \mathcal{W}^n / \sum_i \mathcal{W}^i$ and $c = \mathbb{E}_{\mathbb{M}}[\mathcal{W}]$
Actually,

$$\frac{\mathbb{E}_{\mathbb{M}}[\mathcal{W}\varphi(X)]}{\mathbb{E}_{\mathbb{M}}[\mathcal{W}]} = \mathbb{E}_{\mathbb{Q}}[\varphi(X)]\,.$$

$\mathcal{W}^n$: random weights

- Special case: auto-norm IS; $\mathcal{W}^n = w(X^n)$, $c = \mathbb{M}(w)$, $\mathrm{Var}(\mathcal{W}^n | X^n) = 0$

Algorithm 1 Rejection resampling

Input: $w(x)$ and $C \geq w(x)$
Output: A draw $X \sim \mathbb{Q}$

```
n ← 1, FAIL← TRUE
while FAIL do
    X^n ~ M, U^n ~ U(0,1)
    if U^n C > w(X^n) then
        n ← n + 1
        FAIL←FALSE
    end if
end while
X = X^n
```

- $X^{1:N}$ proposed values
- $\mathcal{W}^n = 0$ to each rejected draw and $\mathcal{W}^n = 1$
- $\mathbb{E}_{\mathbb{M}}[\mathcal{W}^n | X^n] = w(X^n)/[C\mathbb{M}(w)]$
- RS estimator:

$$\frac{\sum_{n=1}^{N} \varphi(X^n)\mathcal{W}^n}{\sum_{n=1}^{N} \mathcal{W}^n}$$

Algorithm 2 Rejection control

Input: N, $w(x)$ and threshold C
Output: Randomly weighted samples $(X^n, \mathcal{W}^n)$ to approximate $\mathbb{Q}$

```
for n = 1 : N do
    X^n ~ M, U^n ~ U(0,1)
    if U^n > 1 ∧ w(X^n)/C then
        W^n = 0
    else
        W^n = w(X^n)/{1 ∧ w(X^n)/C}
    end if
end for
```

- if $w(X^n) > C$ particle maintained with weight $w(X^n)$
- Othws, kept with certain probability and are then equally weighted, with weight C
- $\mathbb{E}[\mathcal{W}^n | X^n = x] = w(x)$
- RS: C upper bound; IS: $C = 0$

Random weights everywhere!

- i-like problems, e.g. inference for stochastic differential equations [3]
- resampling - TBC

[3] Fearnhead, P., Papaspiliopoulos, O., and Roberts, G. O. (2008). Particle filters for partially observed diffusions. *J. R. Stat. Soc. Ser. B Stat. Methodol.*, 70(4):755–777

MSE of RWIS

$$\sqrt{N}\left\{\frac{\sum_{n=1}^{N}\varphi(X^n)\mathcal{W}^n}{\sum_{n=1}^{N}\mathcal{W}^n}-\mathbb{Q}(\varphi)\right\}\Rightarrow\mathcal{N}\left(0,\frac{\mathbb{E}_{\mathbb{M}}[\mathcal{W}^2\overline{\varphi}(X)^2]}{\mathbb{E}_{\mathbb{M}}[\mathcal{W}]^2}\right).$$

But note:

$$\begin{aligned}\frac{\mathbb{E}_{\mathbb{M}}[\mathcal{W}^2]}{\mathbb{E}_{\mathbb{M}}[\mathcal{W}]^2}&=\mathbb{E}_{\mathbb{M}}\left[\mathbb{E}_M\left[\left(\frac{\mathcal{W}}{c}\right)^2\mid X\right]\right]\\&\geq\mathbb{E}_{\mathbb{M}}\left[\mathbb{E}_M\left[\frac{\mathcal{W}}{c}\mid X\right]^2\right]=\frac{\mathbb{M}(w^2)}{\mathbb{M}(w)^2},\end{aligned}$$

Rao-Blackwellization ; why randomise?

Non-negative weights and auxiliary variables

Suppose weights are non-negative. Then particle approximation

$$\mathbb{Q}^N(\mathrm{d}x) := \sum_{n=1}^{N} W^n \delta_{X^n}(\mathrm{d}x), \qquad X^n \sim \mathbb{M}, \qquad W^n = \frac{\mathcal{W}^n}{\sum_{m=1}^{N} \mathcal{W}^m}.$$

Let Z s.t.

$$Z|X = x \sim M(x, \mathrm{d}z) \quad \mathcal{W} = f(Z, X) \quad \text{where} \int f(z, x) M(x, \mathrm{d}z) = c \frac{w(x)}{\mathbb{M}(w)}$$

Define *auxiliary distribution*

$$\mathbb{Q}(\mathrm{d}x, \mathrm{d}z) = \frac{1}{c} f(z, x) \mathbb{M}(\mathrm{d}x) M(x, \mathrm{d}z)$$

Signed weights

Interpretation of weights as unnormalised probabilities is integral part of various algorithms: particle approximation; resampling; Bayes; particle filters. But, often the construction does not ensure this

Methodology for turning a stream of proper weights into non-negative [4]

- Truncate:

$$\mathbb{E}[\{\mathcal{W}^{trunc} - w(x)\}^2 | X = x] \leq \mathbb{E}[\{\mathcal{W} - w(x)\}^2 | X = x]$$

but hard to control approximation bias (no proper weights)

[4] Fearnhead, P., Papaspiliopoulos, O., Roberts, G. O., and Stuart, A. (2010). Random-weight particle filtering of continuous time processes. *J. R. Stat. Soc. Ser. B Stat. Methodol.*, 72(4):497–512

Wald identity and IS

One version: $\mathcal{W}_j$ for $j = 1, 2, \ldots$ have common mean $\mathbb{E}[\mathcal{W}]$, common absolute moment $\mathbb{E}[|\mathcal{W}_j|] = a < \infty$, and T is a stopping time with $\mathbb{E}[T] < \infty$, then

$$\mathbb{E}\left[\sum_{j=1}^{T} \mathcal{W}_j\right] = \mathbb{E}[T]\mathbb{E}[\mathcal{W}]\,;$$

(proof by OST)

Within RWIS: $\mathcal{W}_j^n$ for $j = 1, 2, \ldots$, are independent copies of $\mathcal{W}^n$;

$$S_l^n = \sum_{j=1}^{l} \mathcal{W}_j^n \quad T^n = \min\{l : S_l^n \geq 0\}$$

then provided $\mathbb{E}_{\mathbb{M}}[T^n] < \infty$

$$\mathbb{E}_{\mathbb{M}}[S_{T^n}^n | X^n] = \mathbb{E}_{\mathbb{M}}[T^n] c \frac{w(X^n)}{\mathbb{M}(w)} .$$

Idea in Feanhead et al. is to couple the decision for all particles

Wald RWIS

Theorem

Consider an infinite array of independent random variables $\mathcal{W}_k^n$ *for* $n \in 1:N$ *and* $k = 1, 2, \ldots$ *each with finite fourth moment, and* $\mathbb{E}_{\mathbb{M}}[\mathcal{W}_k^n | X^n = x] = cw(x)/\mathbb{M}(w)$. *We assume that for fixed* n, $\mathcal{W}_k^n$ *are identically distributed for all* k, *with the same distribution as* $\mathcal{W}^n$. *Now define*

$$S_l^n = \sum_{k=1}^{l} \mathcal{W}_k^n,$$

and define the stopping time

$$T = \min\{l : S_l^n \geq 0 \ for\ all\ n \in 1:N\}.$$

Then $\mathbb{E}_{\mathbb{M}}[T] < \infty$, *and*

$$w(X^n)$$

Conditions-complexity

- Fourth moment condition
- Exponential moment condition: $\mathbb{E}_{\mathbb{M}}[T] = \mathcal{O}(\log N)$

CoD

Mantra: importance sampling suffers from the curse of dimensionality

- Yes and potentially very severely:

$$\mathbb{Q}_t(\mathrm{d}x_{0:t}) = \prod_{s=0}^{t} \mathbb{Q}(\mathrm{d}x_s) \quad \mathbb{M}_t(\mathrm{d}x_{0:t}) = \prod_{s=0}^{t} \mathbb{M}(\mathrm{d}x_s) \quad w_t(x_{0:t}) = \frac{\mathrm{d}\mathbb{Q}_t}{\mathrm{d}\mathbb{M}_t}$$

Then

$$\mathrm{Var}_{\mathbb{M}}(w_t) = \mathbb{M}_t(w_t^2) - 1 = r^{t+1} - 1 \quad \text{where } r = \mathbb{M}(w_0^2) \geq 1$$

This is not a (completely) fictitious setting

Universal "curse"

Variance will *always* grow with dimension[5]

Suppose $\mathbb{Q}_s$ and $\mathbb{M}_s$ have densities, q_s and m_s and

$$q_s(x_s|x_{0:s-1}) = \frac{q_s(x_{0:s})}{q_{s-1}(x_{0:s-1})} .$$

Then,

$$w_s(x_{0:s}) = w_{s-1}(x_{0:s-1}) \frac{q_s(x_s|x_{0:s-1})}{m_s(x_s|x_{0:s-1})} ,$$

which indexed by "time" s, is a `martingale`

[5] Kong, A., Liu, J. S., and Wong, W. H. (1994). Sequential imputation and bayesian missing data problems.
J. Am. Statist. Assoc., 89:278–288

Polynomial growth of errors

- $\mathbb{Q}_t$ posterior distribution of θ given t i.i.d observations
- $\mathbb{M}$ prior

Under standard regularity conditions the variance of weights increases as some power of t[6] that depends on the dimension of θ

[6] Chopin, N. (2004). Central limit theorem for sequential Monte Carlo methods and its application to Bayesian inference. *Ann. Statist.*, 32(6):2385–2411

Outline

1 Intro to MC

2 IS
- A first take
- MSE & ESS
- Random Weight IS

3 Introduction to resampling
- Motivating examples
- Resampling framework

Motivation

$$\mathbb{Q}_0^N(\mathrm{d}x_0) = \sum_{n=1}^N W_0^n \delta_{X_0^n}, \qquad X^n \sim \mathbb{M}_0, \qquad W_0^n = \frac{w_0(X_0^n)}{\sum_{m=1}^N w_0(X_0^m)},$$

and now interested in

$$(\mathbb{Q}_0 M_1)(\mathrm{d}x_{0:1}) = \mathbb{Q}_0(\mathrm{d}x_0) M_1(x_0, \mathrm{d}x_1).$$

Two solutions:

SIS

IS from $\mathbb{M}_1 = \mathbb{M}_0 M_1$ to $\mathbb{Q}_0 M_1$ requires:

(a) sample (X_0^n, X_1^n) from $\mathbb{M}_0 M_1$

(b) to compute weights

This ignores the intermediate approximation of $\mathbb{Q}$ by $\mathbb{Q}_0^N$

Resampling

$$\mathbb{Q}_0^N(\mathrm{d}x_0)M_1(x_0,\mathrm{d}x_1) = \sum_{n=1}^N W_0^n M_1(X_0^n,\mathrm{d}x_1)$$

and

$$\frac{1}{N}\sum_{n=1}^N \delta_{\widetilde{X}_{0:1}^n}, \qquad \widetilde{X}_{0:1}^n \sim \mathbb{Q}_0^N(\mathrm{d}x_0)M_1(x_0,\mathrm{d}x_1).$$

One (of several) ways to resample:

$$A_1^{1:N} \sim \mathcal{M}(W_0^{1:N}), \qquad \widetilde{X}_{0:1}^n = (X_0^{A_1^n}, X_1^n), \qquad X_1^n \sim M_1(X_0^{A_1^n},\mathrm{d}x_1)$$

- Recognise some connection to RWIS
- Non-negative weights
- Why resample??

Toy example for the pros

- $\mathcal{X} = \mathbb{R}$, $\mathbb{M}_0$ is $\mathcal{N}(0,1)$, $w_0(x) = \mathbb{1}(|x| > \tau)$, with $\mathbb{M}_0(w_0) = S(\tau) := \mathbb{P}_{\mathbb{M}_0}(|X| > \tau)$
- $\mathbb{Q}_0$ is then a truncated Gaussian distribution
- $M_1(x_0, \mathrm{d}x_1)$ so that $X_1 = \rho X_0 + \sqrt{1-\rho^2} U$, with $U \sim N(0,1)$
- $\varphi(x_1) = x_1$; note that $(\mathbb{Q}_0 M_1)(\varphi) = 0$

$$\widehat{\varphi}_{\mathrm{IS}} = \sum_{n=1}^{N} W_0^n X_1^n \ , (X_0^n, X_1^n) \sim \mathbb{M}_0 M_1 \quad \widehat{\varphi}_{\mathrm{IR}} = N^{-1} \sum_{n=1}^{N} X_1^n \ , X_1^n \sim \mathbb{Q}_0^N M_1$$

$$\sqrt{N}\hat{\varphi}_{\mathrm{IS}} \Rightarrow \mathcal{N}(0, V_{\mathrm{IS}}), \quad \sqrt{N}\hat{\varphi}_{\mathrm{IR}} \Rightarrow \mathcal{N}(0, V_{\mathrm{IR}})$$

with

$$V_{\mathrm{IS}} = \rho^2 \frac{\gamma(\tau)}{S(\tau)} + (1-\rho^2)\frac{1}{S(\tau)}$$

$$V_{\mathrm{IR}} = \underbrace{\rho^2 \frac{\gamma(\tau)}{S(\tau)}}_{\text{IS of } \mathbb{Q}_0} + \underbrace{(1-\rho^2)}_{\text{var of } U} + \underbrace{\rho^2\gamma(\tau)}_{\text{resamp}}$$

where

$$\gamma(\tau) = \frac{2}{\sqrt{\pi}} \frac{\Gamma(3/2, \tau^2)}{S(\tau)}$$

$$V_{\mathrm{IS}} - V_{\mathrm{IR}} = \frac{1}{S(\tau)} - 1 - \rho^2 \left\{ \frac{1}{S(\tau)} - 1 + \gamma(\tau) \right\}$$
$$= -1 + \frac{1}{S(\tau)} \left(1 + \rho^2 - \frac{2\rho^2 \Gamma(3/2, \tau^2)}{\sqrt{\pi}} \right)$$

- Forgetting properties of M_1: for

$$\rho^2 \leq \frac{1 - S(\tau)}{1 - S(\tau) + S(\tau)\gamma(\tau)} = \frac{1 - S(\tau)}{1 - S(\tau) + 2\Gamma(3/2, \tau^2)/\sqrt{\pi}} \in [0, 1]$$

  we have $V_{\mathrm{IS}} \geq V_{\mathrm{IR}}$
- Asymptotics in τ

Resampling as a "safe" option!

Toy example about misuse

- Interest in $\mathbb{Q}_0$
- Suppose weights are equal: $W_0^n = 1/N$.
- Loss of diversity due to resampling?
- probability that no particle will choose a given one as parent is $(1 - 1/N)^N \approx e^{-1} \approx 0.37$ for large N
- Multinomial resampling is particularly bad
- Recall random weights

Resampling sacrifices the past to save the future

Resampling framework: random weights again

Replace

$$\mathbb{Q}^N(\mathrm{d}x) := \sum_{n=1}^{N} W^n \delta_{X^n}(\mathrm{d}x), \qquad X^n \sim \mathbb{M}\,.$$

by

$$\frac{1}{N}\sum_{n=1}^{N} O^n \delta_{X^n}(\mathrm{d}x), \qquad X^n \sim \mathbb{M}$$

where O^n are integer-valued random weights with the unbiased resampling property:

$$\mathbb{E}(O^n) = NW^n\,.$$

Think of O^n as number of off-springs of particle X^n,

$$O^n = \sum_{m=1}^{N} \mathbb{1}(A^m = n) \geq 0$$

Resampling: how?

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Outline

1 How to resample

Inverse CDF

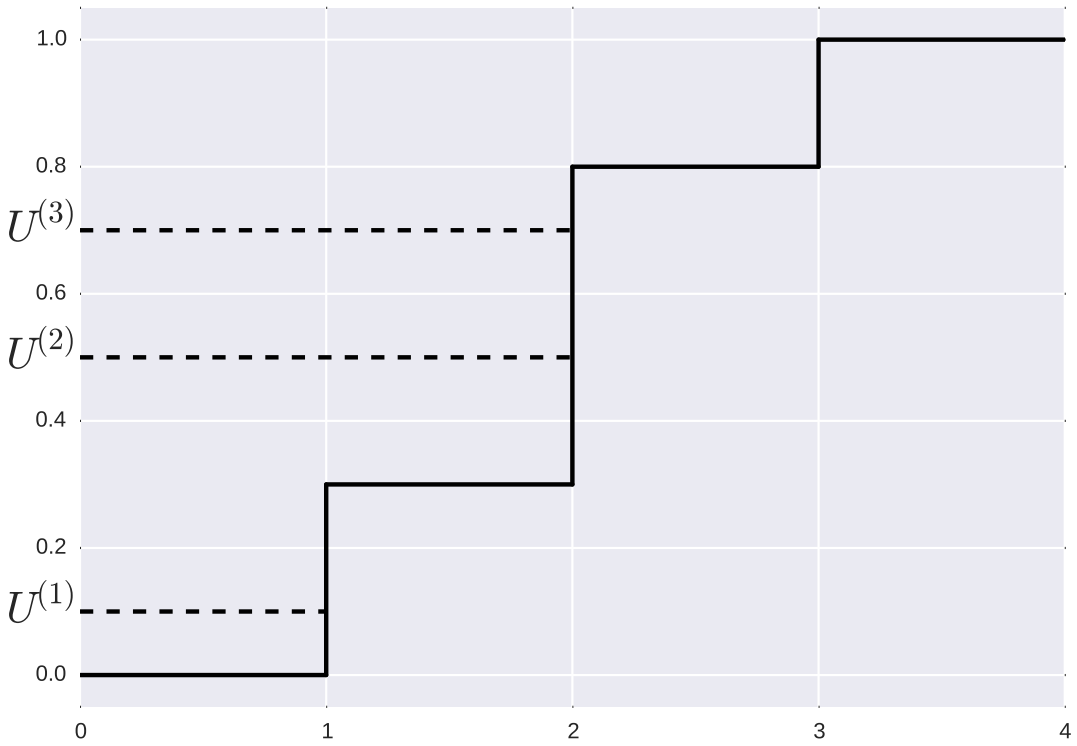

Figure: CDF $F(x) = \sum_{n=1}^{N} W_t^n \mathbb{1}\{n \leq x\}$

Multinomial resampling

In the resampling step, we must simulate N times from $\mathcal{M}(W_t^{1:N})$, the multinomial distribution that generates label n with probability W_t^n.

Multinomial resampling

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Inverse transform method: generate N uniform variates U^m, $m \in 1:N$, and set A_t^m according to:

$$C^{n-1} \leq U^m \leq C^n \quad \Leftrightarrow \quad A_t^m = n$$

where the C^n's are the cumulative weights:

$$C^0 = 0, \quad C^n = C^{n-1} + W_t^n.$$

Multinomial resampling

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where the C^n's are the cumulative weights:

$$C^0 = 0, \quad C^n = C^{n-1} + W_t^n.$$

This suggests resampling costs $\mathcal{O}(N^2)$; however, if we are given uniforms that are already **sorted**, $U^{(1)} < \ldots < U^{(N)}$, then only $2N$ comparisons need to be performed.

Resampling based on sorted uniforms

Algorithm 1:

Normalised weights $W^{1:N}$, and $0 < U^{(1)} < \ldots < U^{(N)} < 1$

$A^{1:N} \in 1 : N$

$s \leftarrow W^1$, $m \leftarrow 1$ **for** $n = 1 \rightarrow N$ **do**

$s < U^{(n)}$ $m \leftarrow m + 1$ $s \leftarrow s + W^m$ $a^n \leftarrow m$

How to generate sorted uniforms

- Generate N uniforms, then sort: $\mathcal{O}(N \log N)$ complexity (not so bad).
- $\mathcal{O}(N)$ complexity by using properties of the Poisson process:

Algorithm 2:

N an ordered sequence $0 < U^{(1)} < \ldots < U^{(N)} < 1$ in $[0,1]$

Sample $E^1, \ldots, E^{N+1} \sim \mathcal{E}(1)$ $C^1 \leftarrow E_1$, $C^n = C^{n-1} + E^n$

(recursively, for $n = 2 : (N+1)$) Return $U^{(n)} = C^n / C^{N+1}$

(for all $n = 1 : N$)

Alternative resampling schemes

We motivated resampling as a way to sample the ancestor X_{t-1}^n from the joint distribution:

$$\sum_{n=1}^{N} W^n \delta_{X_0^n}(\mathrm{d}x_0) M_1(X_0^n, \mathrm{d}x_1)$$

Now imagine $W_{t-1}^n = 1/N$ for all n. The probability of never selecting ancestor X_{t-1}^n is $(1 - 1/N)^N \approx \exp(-1) \approx 0.37$. Seems quite wasteful.

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Now imagine $W_{t-1}^n = 1/N$ for all n. The probability of never selecting ancestor X_{t-1}^n is $(1 - 1/N)^N \approx \exp(-1) \approx 0.37$. Seems quite wasteful.
Let $O^n = \sum_{m=1}^{N} \mathbb{1}\{A^m = n\}$ (number of offsprings). We would like to derive resampling schemes such that

$$\mathbb{E}[O^n] = NW^n$$

while having lower variance than multinomial resampling.

Residual resampling

Let $\mathrm{frac}(x) = x - \lfloor x \rfloor$, and take

$$O^n = \lfloor NW^n \rfloor + \widetilde{O}^n$$

with $\widetilde{O}^n$ taking values in $\mathbb{Z}^+$, and such that $\mathbb{E}[\widetilde{O}^n] = \mathrm{frac}(NW^n)$. To generate the $\widetilde{O}^n$, use multinomial resampling, based on weights $r^n = \mathrm{frac}(NW^n)/R$, with $R = \sum_{n=1}^{N} \mathrm{frac}(NW^n)$.

Residual resampling: the algorithm

Input: normalised weights $W^{1:N}$

Output: $A^{1:N} \in 1:N$

(a) Compute $r^n = \mathrm{frac}(NW^n)$ (for each $n \in 1:N$) and $R = \sum_{n=1}^{N} r^n$.

(b) Construct $A^{1:(N-R)}$ as a vector of size $(N-R)$ that contains $\lfloor NW^n \rfloor$ copies of value n for each $n \in 1:N$.

(c) Sample $A^{N-R+1:N} \sim \mathcal{M}(r^{1:N}/R)$ using multinomial resampling.

Stratified and systematic resampling

We defined multinomial resampling as some operation involving N sorted **IID** uniforms $U^{(n)}$. Taking instead

$$U^{(n)} \sim \mathcal{U}\left(\frac{n-1}{N}, \frac{n}{N}\right)$$

gives **stratified resampling**. Reducing randomness further, taking

$$U^{(n)} = (n - 1 + U)/N$$

(based on a *single* uniform $U \sim \mathcal{U}(0,1)$) leads to **systematic resampling**.

Particle filtering

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Outline

1. Objectives
2. The algorithm
3. Particle algorithms for a given state-space model
4. When to resample?

Objectives

- introduce a generic PF algorithm for a given Feynman-Kac model;
- discuss the different algorithms one may obtain for a given state-space model, by using different Feynman-Kac formalisms;
- give more details on the implementation, complexity, and so on of the algorithm.

Outline

1 Objectives

2 The algorithm

3 Particle algorithms for a given state-space model

4 When to resample?

Input

- A Feynman-Kac model such that:
  - the weight function G_t may be evaluated pointwise (for all t);
  - it is possible to simulate from $\mathbb{M}_0(\mathrm{d}x_0)$ and from $M_t(x_{t-1}, \mathrm{d}x_t)$ (for any x_{t-1} and t)

Input

- A Feynman-Kac model such that:
  - the weight function G_t may be evaluated pointwise (for all t);
  - it is possible to simulate from $\mathbb{M}_0(\mathrm{d}x_0)$ and from $M_t(x_{t-1}, \mathrm{d}x_t)$ (for any x_{t-1} and t)
- The number of particles N

Generic SMC algorithm

All operations to be performed for all $n \in 1:N$.
At time 0:

(a) Generate $X_0^n \sim \mathbb{M}_0(\mathrm{d}x_0)$.

(b) Compute $w_0^n = G_0(X_0^n)$, $W_0^n = w_0^n / \sum_{m=1}^N w_0^m$, and $L_0^N = N^{-1} \sum_{n=1}^N w_0^n$.

Recursively, for $t = 1, \ldots, T$:

(a) Generate ancestor variables $A_t^n \in 1:N$ independently from $\mathcal{M}(W_{t-1}^{1:N})$.

(b) Generate $X_t^n \sim M_t(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$.

(c) Compute $w_t^n = G_t(X_{t-1}^{A_t^n}, X_t^n)$, $W_t^n = w_t^n / \sum_{m=1}^N w_t^m$, and $L_t^N = L_{t-1}^N \{N^{-1} \sum_{n=1}^N w_t^n\}$.

Output

The algorithm delivers the following approximations at each time t:

$$\frac{1}{N}\sum_{n=1}^{N}\delta_{X_t^n} \quad \text{approximates } \mathbb{Q}_{t-1}(\mathrm{d}x_t)$$

$$\mathbb{Q}_t^N(\mathrm{d}x_t) = \sum_{n=1}^{N} W_t^n \delta_{X_t^n} \quad \text{approximates } \mathbb{Q}_t(\mathrm{d}x_t)$$

$$L_t^N \quad \text{approximates } L_t$$

some comments

- by *approximates*, we mean: for any test function φ, the quantity

$$\mathbb{Q}_t^N(\varphi) = \sum_{n=1}^{N} W_t^n \varphi(X_t^n)$$

converges to $\mathbb{Q}_t(\varphi)$ as $N \to +\infty$ (at the standard Monte Carlo rate $\mathcal{O}_P(N^{-1/2})$).

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- complexity is $\mathcal{O}(N)$ per time step.

Outline

1. Objectives
2. The algorithm
3. Particle algorithms for a given state-space model
4. When to resample?

Principle

We now consider a given state-space model:

- with initial law $P_0(\mathrm{d}x_0)$ and Markov kernel $P_t(x_{t-1}, \mathrm{d}x_t)$ for $\{X_t\}$;
- with conditional probability density $f_t(y_t|x_t)$ for $Y_t|X_t$

and discuss how the choice of a particular Feynman-Kac formalism leads to more or less efficient particle algorithms.

The bootstrap filter

Bootstrap Feynman-Kac formalism:

$$M_t(x_{t-1}, \mathrm{d}x_t) = P_t(x_{t-1}, \mathrm{d}x_t), \quad G_t(x_{t-1}, x_t) = f_t(y_t|x_t)$$

then $\mathbb{Q}_t$ is the filtering distribution, L_t is the likelihood of $y_{0:t}$, and so on.

The resulting algorithm is called the **bootstrap filter**, and is particularly simple to interpret: we sample particles from Markov transition $P_t(x_{t-1}, \mathrm{d}x_t)$, and we reweight particles according to how compatible they are with the data.

The bootstrap filter: algorithm

All operations to be performed for all $n \in 1:N$.
At time 0:

(a) Generate $X_0^n \sim P_0(\mathrm{d}x_0)$.

(b) Compute $w_0^n = f_0(y_0|X_0^n)$, $W_0^n = w_0^n / \sum_{m=1}^N w_0^m$, and $L_0^N = N^{-1} \sum_{n=1}^N w_0^n$.

Recursively, for $t = 1, \ldots, T$:

(a) Generate ancestor variables $A_t^n \in 1:N$ independently from $\mathcal{M}(W_{t-1}^{1:N})$.

(b) Generate $X_t^n \sim P_t(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$.

(c) Compute $w_t^n = f_t(y_t|X_t^n)$, $W_t^n = w_t^n / \sum_{m=1}^N w_t^m$, and $L_t^N = L_{t-1}^N \{N^{-1} \sum_{n=1}^N w_t^n\}$.

The bootstrap filter: output

$$\frac{1}{N}\sum_{n=1}^{N}\varphi(X_t^n) \quad \text{approximates } \mathbb{E}[\varphi(X_t)|Y_{0:t-1}=y_{0:t-1}]$$

$$\sum_{n=1}^{N}W_t^n\varphi(X_t^n) \quad \text{approximates } \mathbb{E}[\varphi(X_t)|Y_{0:t}=y_{0:t}]$$

$$L_t^N \quad \text{approximates } p(y_{0:t})$$

The bootstrap filter: pros and cons

Pros:

- particularly simple
- does not require to compute the density $X_t|X_{t-1}$: we can apply it to models with **intractable dynamics**

Cons:

- We simulate particles *blindly*: if $Y_t|X_t$ is very informative, few particles will get a non-negligible weight.

The guided PF

Guided Feynman-Kac formalism: M_t is a user-chosen **proposal** kernel such that $M_t(x_{t-1}, \mathrm{d}x_t)$ dominates $P_t(x_{t-1}, \mathrm{d}x_t)$, and

$$G_t(x_{t-1}, x_t) = \frac{f_t(y_t|x_t)P_t(x_{t-1}, \mathrm{d}x_t)}{M_t(x_{t-1}, \mathrm{d}x_t)} = \frac{f_t(y_t|x_t)p_t(x_t|x_{t-1})}{m_t(x_t|x_{t-1})}$$

(assuming in the second line that both kernels admit a density wrt a common measure). We still have that $\mathbb{Q}_t(\mathrm{d}x_t)$ is the filtering distribution, and L_t is the likelihood.
We call the resulting algorithm the **guided particle filter**, as in practice we would like to choose M_t so as to **guide** particles to regions of high likelihood.

The guided PF: choice of M_t (local optimality)

Suppose that (G_s, M_s) have been chosen to satisfy (**??**) for $s \leq t-1$. Among all pairs (M_t, G_t) that satisfy (**??**), the Markov kernel

$$M_t^{\mathrm{opt}}(x_{t-1}, \mathrm{d}x_t) = \frac{f_t(y_t|x_t)}{\int_{\mathcal{X}} f(y_t|x')\, P_t(x_{t-1}, \mathrm{d}x')} P_t(x_{t-1}, \mathrm{d}x_t)$$

minimises the variance of the weights, $\mathrm{Var}\left[G_t(X_{t-1}^{A_t^n}, X_t^n)\right]$.

Interpretation and discussion of this result

- M_t^{opt} is simply the law of X_t given X_{t-1} and Y_t. In a sense it is the perfect compromise between the information brought by $P_t(x_{t-1}, \mathrm{d}x_t)$ and by $f_t(y_t|x_t)$.
- In most practical cases, M_t^{opt} is not tractable, hence this result is mostly indicative (on how to choose M_t).
- Note also that the local optimality criterion is debatable. For instance, we do not consider the effect of *future* datapoints.

A first example: stochastic volatility

There, the log-density of $X_t|X_{t-1}, Y_t$ is (up to a constant):

$$-\frac{1}{2\sigma^2}\left\{x_t - \mu - \phi(x_{t-1} - \mu)\right\}^2 - \frac{x_t}{2} - \frac{e^{-x_t}}{2}y_t^2$$

We can use $e^{x-x_0} \approx 1 + (x - x_0) + (x - x_0)^2/2$ to get a Gaussian approximation.

A second example: bearings-only tracking

In that case, $P_t(x_{t-1}, \mathrm{d}x_t)$ imposes deterministic constraints:

$$X_t(k) = X_{t-1}(k) + X_{t-1}(k+2), \quad k = 1, 2$$

We can choose a M_t that imposes the same constraints. However, in this case, we find that

$$M_t^{\mathrm{opt}}(x_{t-1}, \mathrm{d}x_t) = P_t(x_{t-1}, \mathrm{d}x_t).$$

Discuss.

Guided particle filter pros and cons

Pro:

- may work much better that bootstrap filter when $Y_t|X_t$ is informative (provided we are able to derive a good proposal).

Cons:

- requires to be able to compute density $p_t(x_t|x_{t-1})$.
- sometimes local optimality criterion is not so sound.

The auxiliary particle filter

In the auxiliary Feynman-Kac formalism, an extra degree of freedom is gained by introducing **auxiliary** function η_t, and set:

$$G_0(x_0) = f_0(y_0|x_0)\frac{P_0(\mathrm{d}x_0)}{M_0(\mathrm{d}x_0)}\eta_0(x_0),$$

$$G_t(x_{t-1}, x_t) = f_t(y_t|x_t)\frac{P_t(x_{t-1}, \mathrm{d}x_t)}{M_t(x_{t-1}, \mathrm{d}x_t)}\frac{\eta_t(x_t)}{\eta_{t-1}(x_{t-1})}\,.$$

so that

$$\mathbb{Q}_t(\mathrm{d}x_{0:t}) \propto \mathbb{P}(\mathrm{d}x_{0:t}|Y_{0:t} = y_{0:t})\eta_t(x_t)$$

and we recover the filtering distribution by reweighting by $1/\eta_t$.
Idea: choose η_t so that G_t is as constant as possible.

Output of APF

Let $\tilde{w}_t^n := w_t^n / \eta_t(X_t^n)$, $\tilde{W}_t^n := \tilde{w}_t^n / \sum_{m=1}^N \tilde{w}_t^m$, then

$$\frac{1}{\sum_{m=1}^N \frac{\tilde{W}_t^m}{f(y_t|X_t^m)}} \sum_{n=1}^N \frac{\tilde{W}_t^n}{f_t(y_t|X_t^n)} \varphi(X_t^n) \quad \text{approx. } \mathbb{E}[\varphi(X_t)|Y_{0:t-1} = y_{0:t-1}]$$

$$\sum_{n=1}^N \tilde{W}_t^n \varphi(X_t^n) \quad \text{approx. } \mathbb{E}[\varphi(X_t)|Y_{0:t} = y_{0:t}]$$

$$L_t^N \times N^{-1} \sum_{n=1}^N \tilde{w}_t^n \quad \text{approx. } p(y_{0:t})$$

Local optimality for M_t and η_t

For a given state-space model, suppose that (G_s, M_s) have been chosen to satisfy (**??**) for $s \leq t-2$, and M_{t-1} has also been chosen. Among all pairs (M_t, G_t) that satisfy (**??**) and functions η_{t-1}, the Markov kernel

$$M_t^{\mathrm{opt}}(x_{t-1}, \mathrm{d}x_t) = \frac{f_t(y_t|x_t)}{\int_{\mathcal{X}} f(y_t|x')\, P_t(x_{t-1}, \mathrm{d}x')} P_t(x_{t-1}, \mathrm{d}x_t)$$

and the function

$$\eta_{t-1}^{\mathrm{opt}}(x_{t-1}) = \int_{\mathcal{X}} f(y_t|x')\, P_t(x_{t-1}, \mathrm{d}x')$$

minimise $\mathrm{Var}\left[G_t(X_{t-1}^{A_t^n}, X_t^n)/\eta_t(X_t^n)\right]$.

Interpretation and discussion

- We find again that the optimal proposal is the law of X_t given X_{t-1} and Y_t. In addition, the optimal auxiliary function is the probability density of Y_t given X_{t-1}.
- For this ideal algorithm, we would have

$$G_t(x_{t-1}, x_t) = \eta_t^{\mathrm{opt}}(x_t);$$

the density of Y_{t+1} given $X_t = x_t$; not constant, but intuitively less variable than $f_t(y_t|x_t)$ (as in the bootstrap filter).

Example: stochastic volatility

We use the same ideas as for the guided PF: Taylor expansion of log-density, then we integrate wrt x_t.

APF pros and cons

Pros:

- usually gives some extra performance.

Cons:

- a bit difficult to interpret and use;
- they are some (contrived) examples where the auxiliary particle filter actually performs worse than the bootstrap filter.

Note on the generality of APF

From the previous descriptions, we see that:

- the guided PF is a particular instance of the auxiliary particle filter (take $\eta_t = 1$);
- the bootstrap filter is a particular instance of the guided PF(take $M_t = P_t$).

This is why some recent papers focus on the APF.

Which resampling to use in practice?

- Systematic resampling is fast, easy to implement, and seems to work best; but no supporting theory.
- We **do** have some theoretical results regarding the fact that multinomial resampling is dominated by most other resampling schemes. (So don't use it!)
- On the other hand, multinomial resampling is easier to study formally (because again it is based on IID sampling).

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Resampling or not resampling, that is the question

For the moment, we resample every time. When we introduced resampling, we explained that the decision to resample was based on a trade-off: adding noise at time $t-1$, while hopefully reducing noise at time t (assuming that $\{X_t\}$ forgets its past).

Resampling or not resampling, that is the question

For the moment, we resample every time. When we introduced resampling, we explained that the decision to resample was based on a trade-off: adding noise at time $t-1$, while hopefully reducing noise at time t (assuming that $\{X_t\}$ forgets its past).
We do know that never resample would be a bad idea: consider $M_t(x_{t-1}, \mathrm{d}x_t)$ defined such that the X_t are IID $\mathcal{N}(0,1)$, $G_t(x_t) = \mathbb{1}(x_t > 0)$. (More generally, recall the curse of dimensionality of importance sampling.)

The ESS recipe

Trigger the resampling step whenever the variability of the weights is too large, as measured by e.g. the ESS (effective sample size):

$$\mathrm{ESS}(W_t^{1:N}) := \frac{1}{\sum_{n=1}^N (W_t^n)^2} = \frac{\{\sum_{n=1}^N w_t(X^n)\}^2}{\sum_{n=1}^N w_t(X^n)^2}.$$

Recall that $\mathrm{ESS}(W_t^{1:N}) \in [1, N]$, and that if k weights equal one, and $N - k$ weights equal zero, then $\mathrm{ESS}(W_t^{1:N}) = k$.

PF with adaptive resampling

(Same operations at $t = 0$.)
Recursively, for $t = 1, \ldots, T$:

(a) **If** $\mathrm{ESS}(W_{t-1}^{1:N}) < \gamma N$
generate ancestor variables $A_{t-1}^{1:N}$ from resampling distribution $\mathcal{RS}(W_{t-1}^{1:N})$, and set $\hat{W}_{t-1}^n = W_{t-1}^{A_t^n}$;
Else (no resampling)
set $A_{t-1}^n = n$ and $\hat{W}_{t-1}^n = 1/N$

(b) Generate $X_t^n \sim M_t(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$.

(c) Compute $w_t^n = (N\hat{W}_{t-1}^n) \times G_t(X_{t-1}^{A_t^n}, X_t^n)$,
$L_t^N = L_{t-1}^N \{N^{-1} \sum_{n=1}^N w_t^n\}$, $W_t^n = w_t^n / \sum_{m=1}^N w_t^m$.